Suicide rates in India and its implications for National Mental Health Programme

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Abstract:

According to World Health Organization (WHO) suicide is a global burden and it is the third leading cause of death in the world. India accounts for a third largest in the world for female suicides yearly and nearly a fourth largest for male suicides. The India's share for suicides rate significantly rising in global share from 1990. Mental issues are one of the main reasons of fatality in India, yet a precise comprehension of their pervasiveness, also disease load and risk factors is not promptly accessible for each territory of India. By using NCRB and Health Management Information System (HMIS) data for the time period of 2010-2019 the major objective is to present a research to analyze the trend and pattern of suicides deaths in India and to study the Suicide Growth Rate in the country in the last one decade. The suicide trend comparison shows a contradicting result between HMIS and NCRB data. HMIS show increasing trend for last 9 years and NCRB has shown a declining trend. It is due to the limitation of HMIS suicide data collection. HMIS collects data only from the public health facility level but NCRB has various resources for collection of data. The major reasons of the suicides are family problems, illness, other causes, causes not known, drug abuse/ addiction and bankruptcy or sudden change in economic position which accounts for more than 80% of suicides. The maximum numbers of deaths are reported in the young adolescent cohorts. Another purpose of the study is to critically review India's Mental Health Programme (NMHP) and District Mental Health Programme (DMHP) and its implication with respect to suicide deaths in India. In this connection, it may be suggested that the District Hospitals and Hospital for Mental Health can be strengthened by collaborative care model (community-based care and facility-based care services) under district mental health program with a tele-psychiatric mobile van, adequate financial resource, and training. PHCs can be equipped with essential psychotropic drugs and teleconferencing or video conferencing for specialist consultation by psychologists and psychiatrists on a fortnightly or monthly basis from district or state level team.

Keywords: *Mental Health, Suicide Rates, National Mental Health Programmes (NMHP), District Mental Health Programmes (DMHP), National Crime Records Bureau (NCRB), Health Management Information System (HMIS)etc..*

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1. Introduction

1.1. Framing the problem

The definition of mental health given by World Health Organization (WHO) is defined as: "*a state of wellbeing in which every individual realizes his/her own potential, can cope with the normal stresses of life, can work productively and fruitfully, and is able to make a contribution to her or his community".* Any disturbance to these states has a severe impact on the overall health of a person which takes into account emotional, psychological and social-wellbeing leading to welfare losses at personal and societal level. 13% of mental disorders have been reported in 2004 as a global disease burden. The WHO for the first time endorses a complete mental health action plan (MHAP) in 2013–2020, which has called for change in perspectives of attitudes, extension of services, and efficient utilization of scarce resources in promoting mental prosperity, preventing mental issues, and shielding privileges of individuals experiencing mental sicknesses. It is concentrated on four key destinations:"*to reinforce compelling initiative and administration for psychological well-being; give complete, coordinated, and responsive emotional well-being and social consideration administrations in network-based settings; execute procedures for advancement and avoidance in emotional wellness; and fortify data frameworks, proof, and exploration for emotional well-being*".

Every year on an average 8 lakhs people commit suicide around the globe. For every person who commit suicides there are many more people who attempt suicide every year. A prior suicide attempt is the absolute factor for suicide in manner like populace. There are indications that for each adult who died of suicide there may have been more than 20 others attempting suicide. Suicide is the third most important cause of deaths in 15 to 19 years old population. Around 79% of the suicide occurs in low-middle income group countries in 2016. Ingestion of pesticides, hanging and firearms are among the most common methods of suicides around the world (WHO 2019).

While the co-relation among suicide and mental issues is observed in nations with high-income, numerous suicides occur hastily in snapshots of emergency with a breakdown in the ability to manage life stresses, for example, monetary issues, relationship separation or constant pain and sickness. Furthermore, conflicts which have experience, catastrophic issues, brutality, misuse, or misfortune and a feeling of separation are unequivocally connected with suicide conduct. Suicide rates are additionally high among feeble social gatherings who experience segregation, for example, exiles and migrants; indigenous people groups; lesbian, gay, indiscriminate, transsexual, intersex (LGBTI) people; and detainees. As far the most grounded hazardous factor for self-destruction is past suicide attempt.

It is observed that "Stigma" has been surrounded by mental disorders and suicides meaning that people who take their own life or who attempt to suicide are not taking any help and thus they are not getting the desired help they need. Suicide can be prevented but the majority of cases are not

addressed properly due to the lack of awareness on this aspect as a public health issue. Besides that, suicide considered to be a taboo in many societies and is not been discussed out in open. As per the records, only 38 countries have included the national suicide prevention strategy. Thus, the community awareness and taboo break down is a vital component for countries to make a progress in the prevention of suicides

The availability and reliability of suicide data are quite poor in most of the countries. The issue with the poor quality data lies in fact that there is under reporting and no proper bifurcation is mentioned as to why suicides are happening. This study uses a novel approach to compare suicide rates from different sources and give holistic information of available suicide databases.

1.2. The Indian Context

1.2.1 Mental Health: An Emerging Science

Mental health problems issues are a reason for incredible worry in India according to the WHO insights (2017) and the ongoing study (2015 - 2016) led by the Bengaluru-based National Institute of Mental Health and Neuro Sciences (NIMHANS) in 12 states, covering 34,802 individuals lends interesting insights. The NIMHANS study uncovers 10 % of the populace has normal mental issues and 1.9 % of the populace experiences extreme mental issues. Schizophrenia, bipolar emotional issue (BPAD), sadness, uneasiness issues, psychoses, fear, self destruction, disposition issues, hypochondriac or stress related issues, post horrible pressure issue, conjugal disharmony, rest issues, liquor reliance and substance abuse, and dementia are turning out to be normal issues in everybody. The report cautioned that with urbanization, the weight of emotional wellness issues on the general public is relied upon to increment. The pervasiveness of mental bleakness, an upsetting situation, the report proposes could be because of relentless ways of life, encountering pressure, complexities of living, a breakdown of emotionally supportive networks and difficulties of financial precariousness (Shankardass, 2018). WHO in an ongoing report (2017) suggests prompt master mediation since emotional well-being insights in the nation may ascend and influence around 20 % of the populace by 2020 from 2015 gauge of generally 7.5 % of India's populace experiencing some type of psychological sickness (major or minor) (WHO, 2017).

The NIMHANS report shows lifetime predominance of mental issue as 13.7 % with 10.6 % of them requiring prompt intercessions. Extrapolating the after effects of the 12 states to whole nation would mean at any rate 150 million Indians needs intercession for emotional well-being. This National Mental Health study demonstrates urban regions to be generally influenced, the issues being 2–3 times higher in Metro cities. It is seen that, around 1 out of 20 experiences of the ill effects of depression, being higher in females in the age-group of 40-49 years. 22.4 % of the population over 18 years experiences substance use issue, with the most elevated contributed by tobacco and liquor use issue and identified more among males. While predominance of psychological sickness is higher

among males (13.9 %) in comparison to females (7.5 %), particular dysfunctional behaviours like mind-set issues (despondency, psychotic issues, phobic uneasiness issues and so on) are more in females. Mental issues and stress related disease is additionally observed to be more in ladies. Commonness in youngsters matured somewhere in the range of 13 to17 years apparently is 7.3 % (NIMHANS, 2016).

The growing incidence and predominance of mental issues as the report recommends at present just 30 million Indians approach psychological wellness office. A few examinations demonstrate that only 10 % of Indians with psychological well-being issues get proof-based medicines. Treatment gaps exists more than 70 % because of inadequate financing of mental, neurological, and substance use disorders. India's spending on emotional wellness care has reliably been deficient despite the fact that the quantity of people influenced by dysfunctional behaviour is huge. Despite the huge weight of psychological maladjustment in 2011 just 0.06 % of 4.16 % of the GDP on wellbeing in the nation was assigned at a public level for outpatient mental consideration. India's number of psychological well-being beds was discovered to be well beneath normal with just 2.15 beds per 100,000 contrasted with the worldwide figure of 6.5 beds per 100,000. Therefore, a significant worry in the nation is that in spite of three out of four people encountering extreme mental issues, there are gigantic hole in treatment. Aside from epilepsy, the treatment hole for all psychological problems is more than 60 %. Actually, the monetary weight of mental issues is immense to the point that influenced families need to spend almost 1,000 - 1,500 rupees per month especially for the treatment and to access care (Shankardass, 2017).

India was one of the first nations in creating world to receive a National Mental Health Program (NMHP) in 1982. A progression of difficulties and execution of disappointments, all things considered, prompted it’s under performance. However, interest of multiple stakeholders got reignited after the WHO called for scaling up action against mental illnesses in 2013. Since then, India has been witnessing a great push for mental health with revamping of its MHP, framing of its first mental health policy fully in line with human rights covenants, and enactment of mental health legislation. Underlying principles of newly rolled out District MHP (DMHP) was based on six key perspectives, namely life course, recovery, equity, evidence based, health systems, and right based.

As a matter of first importance explanation behind India to lose its mental health is the absence of mindfulness and affectability about the issue. It is considered to be a stigma for people who suffer from this disorder which are eventually tagged as Lunatics by the people of the society. Leading to a vicious cycle of shame, suffering and isolation of the patients. There is a manpower crunch in the area of mental healthcare in India. According to WHO, in 2011, there were 0·301 psychiatrists and 0·047 psychologists for every 100,000 patients suffering from a mental health disorder in India. These issues aggravate the scale of the problem and need immediate redressal. At present, India layout 1.3% of its

complete financial plan on mental health. This is horribly less when we compare with other countries and an intermediary pointer of low need on wellbeing plan.

1.3. Suicides in India

Suicide is the third driving reason for death among youth around the world. There is a developing acknowledgment in the avoidance techniques which should be custom-made to the area explicit socioeconomics of a nation and to be executed in a socially delicate way. There has been an expansion in the paces of suicide rates in India throughout the years, in spite of the fact that patterns of the increments and decrements in these rates have been available. Suicide is among the three major explanations behind death among youth around the globe. All things considered approximately one million people die from suicide and 20 times more people attempt suicide; a global mortality rate of 16 per 100,000, or one death every 40 seconds and one attempt every 3 seconds, on an average. Suicide overall was evaluated to speak to 1.8% of the absolute worldwide weight of illness in 1998; in 2020, this figure is anticipated to be 2.4% in nations with market and previous communist economies (WHO 2012). In ancient India, suicide to a great extent has been impacted by several motives which include the sake of honour, religious, and socio-cultural beliefs apart from mental and different causes. Ramayana and Mahabharata are the twin legends of India and down hundreds of years they have impacted the musings, temper, direct, and culture of our kin. In Mahabharata, on hearing the news about the death of his son Abhimanyu in the battle, Arjuna wanted a huge fire to be prepared for him to commit suicide; however, that was prevented by Lord Krishna by reminding Arjuna about his own earlier advice to another old man who tried to commit suicide by falling into fire on the death of his son (Rao, 1986).

Suicide is a significant issue in the context of India. More than one lakh (100,000) lives are lost every year due to suicide in our country. Over the most recent twenty years, the suicide has been expanded from 7.9 to 10.3 per 100,000. There is a wide assortment in the suicide rates inside the country. The southern regions, Kerala, Karnataka, Andhra Pradesh and Tamil Nadu have a suicide rate is greater than 15 % while in the Northern regions, Punjab, Uttar Pradesh, Bihar and Jammu and Kashmir, the suicide rate is less than 3 %. This variable pattern has been consistent throughout the previous twenty years. Higher proficiency in education levels, a dependable and better framework of reporting, lower outside hostility, higher financial status and better standards are the potential avocations for the higher suicide rate in the southern regions of India (Vijaykumar, 2007).

In India the majority of the suicide rates are committed by those who are less than 30 years. Below the age of 44 years, 71 % of suicides are being committed in India which has a great impact on social, emotional, and economic burden on our society. In the western counterparts there has been a narrow male to female ratio of 1.3:1 referring those women commits suicide more than men (Mayer & Ziaian, 2002). The common methods to commit suicide are Poisoning (36.6%), hanging (32.1%) and self-

immolation (7.9%). Two enormous epidemiological verbal autopsy examination concentrates in provincial regions of Tamil Nadu which uncover that the yearly suicide rate is 6 - 9 times the official rate (NCRB, 2005).It is assessed that one in 60 persons in our country are affected by suicide. It incorporates both, those who have endeavoured suicide and those who have been affected by the suicide of a close family or friend. Thus, suicide is a significant public and mental health problem, which requests urgent action (Vijaykumar, 2007).

Imperfect market conditions and crash in agricultural crop prices is a leading cause of suicides among male farmers in rural India (Nagaraj, 2008). The failure of Bt cotton contribute to farmer indebtedness leading to suicides. Its failure is mostly in terms of the environmental conditions in which it is sown (Gruere, et al., 2008).

The WHO has defined sustainable development goals and elaborated the impact of mental illnesses and suicide on them. The suicide rate in India in 2015 at 15.7/100,000 is higher than the regional average of 12.9 and the global average of 10.6 (Koshy, 2015).The way to deal with understanding suicide must be multidisciplinary, including psychologists, psychiatrists, toxicologists, physiologists and doctors, since suicide is a multi-faceted and complex occasion. The untimely and rough passing of the casualties has negative repercussions in the public eye and ought to be forestalled at whatever point conceivable (Kumar et al., 2013).

1.4. Objectives

In light of the above problem the present study is outlined in such a way to attain the following five main objectives which are as follows:

1. To analyze the trend and pattern of suicides deaths in India as well as to study the Suicide Growth Rate in the country in last one decade.
2. To give a comparative analysis between HMIS and NCRB data and ranking based on the number of suicides.
3. To assess the heterogeneity and trends of suicide across the various states of India.
4. To analyse the demographic variation and major reasons of suicides from the 2010 to 2017 of NCRB data.
5. To assess the implication and innovation of NMHP with respect to suicides in India.

1.5. Data Sources & Methods

The data for the present study is based on secondary data and it is extracted from the National Crime Record Bureau (NCRB) and Health Management Information System (HMIS) for the time period from 2010-11 to 2019-20 respectively. The NCRB database formed out of Bureau of Police Research and Development is the principal source of information on connected issues about crimes, accidental deaths, suicides and prison statistics. HMIS is a web-based information system that has been put forth by the MoHFW, GOI to keep a track of all the programme interventions and furnish the required

inputs for policy formulations. Therefore, the suicide numbers will give us a clear picture of India so as to better understand the concern behind mental illness and which states have made to the top 5 rank in which the suicidal rates have increased at an alarming rate. thus fulfilling the desired objectives which have been outlined above. Standard statistical methods such as computing growth rates, cross-tabulations, etc. has been used to formulate the findings.

2. Suicides: Insights from long-term surveillance data

2.1. Prevalence of suicides in India: Evidence from HMIS

India's suicide rates have drastically increased in the last one decade as per the HMIS data (see figure 1). As per the National Crime Record Bureau (NCRB) approximately one lakh people died every year due to suicide in India. In this section the number of suicide data was collected from HMIS and depicted in figure 1. The HMIS suicide data was only collected at the health facilities all across India so HMIS did not have the complete data of suicides. In last ten years the total number of suicides recorded in HMIS have significantly increased.

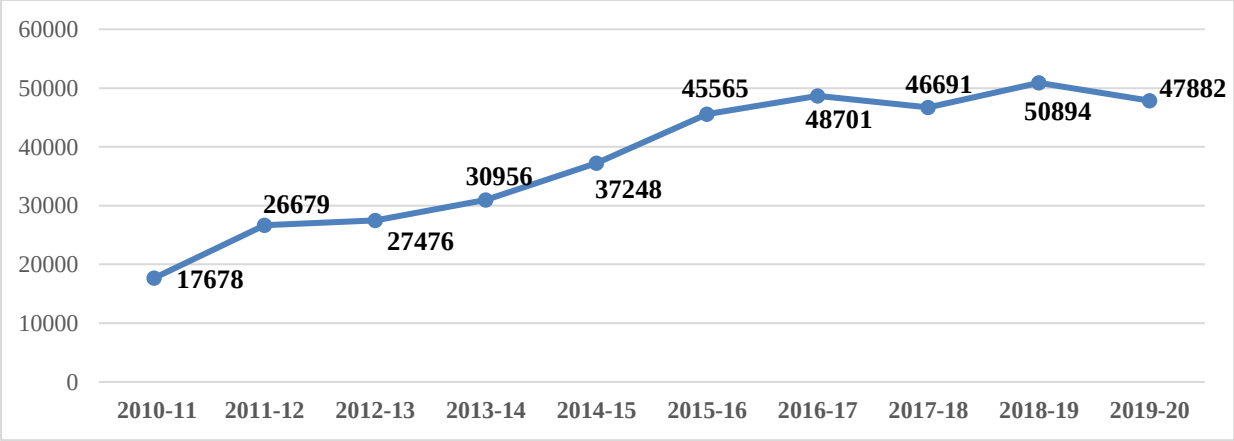

Figure 1: Number of suicides (6+ age group) in India in last one decade 2010-2019 (HMIS)

In 2010, total numbers of suicides were 17678 and in year 2019 it was recorded as 47882. More than 60 % of the suicides were committed in Tami Nadu, Maharashtra, Karnataka, Andhra Pradesh, Madhya Pradesh and West Bengal. In the last one decade the above-mentioned states are having the highest number of suicides among all other Indian states.

2.2. Annual Suicide Growth Rate

The states and UTs that have reported significant increase in suicides in 2011 over 2010 were Maharashtra (from 149 in 2010 to 3153 in 2011 i.e. an increase of 2016%) followed by Chhattisgarh (1392%), Daman & Diu (200%), Bihar (106%), Punjab (112%), Assam (79%), Orissa (67%), Dadar Nagar Haweli (58%) and Haryana (57%) while highest number of decrease was reported in Sikkim (from 22 in 2010 to 7 in 2011) (Figure 2).

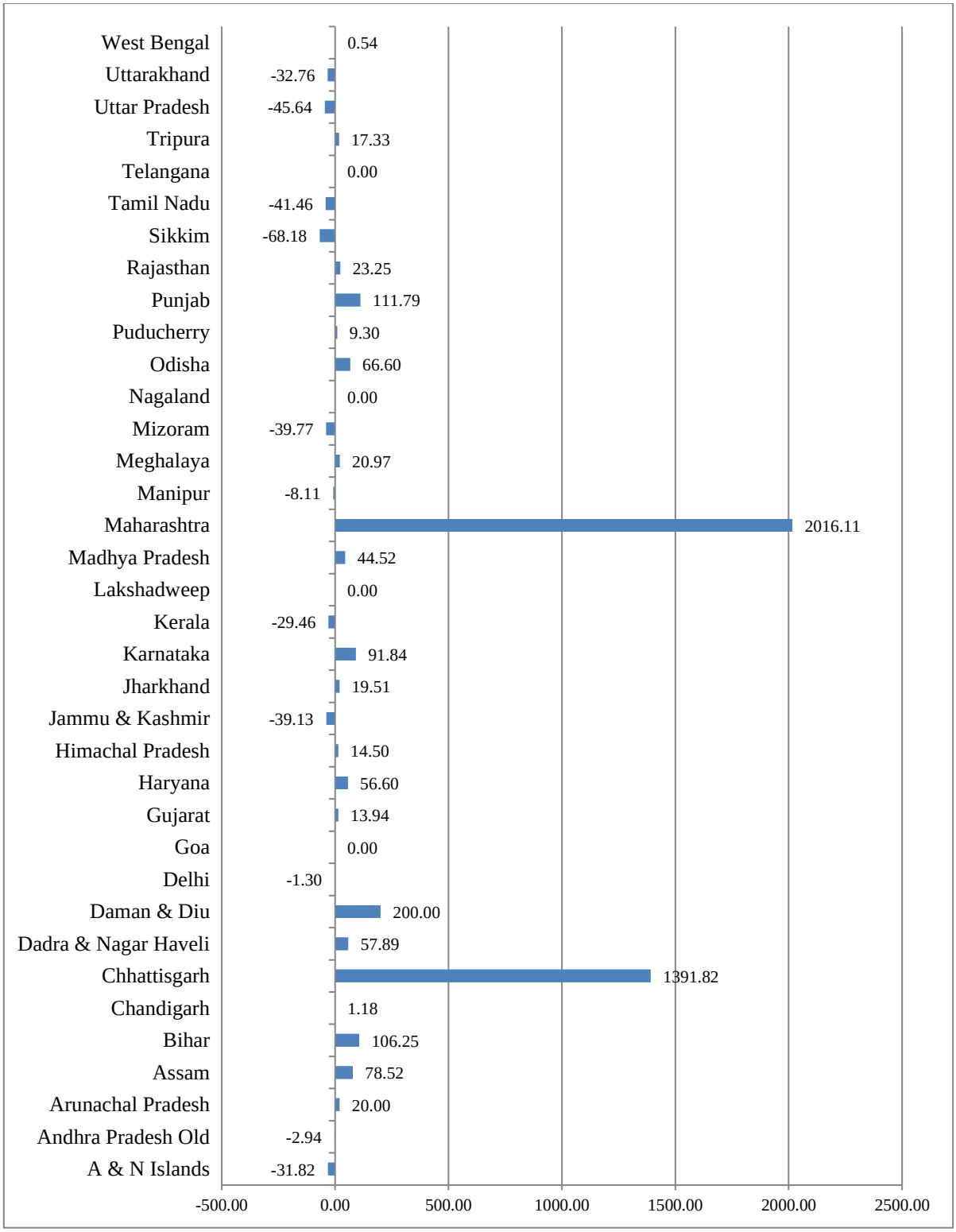

Figure 2: Suicide deaths growth rate, 2011 over 2010, HMIS

Similarly, in 2017 over 2016, the suicide growth rate increased for Daman & Diu (100%), Manipur (56%), Andaman & Nicobar (41%), Haryana (24%), Chandigarh (22%) and Telangana (19%) while highest percentage decrease was reported in Lakshadweep (100%), Arunachal Pradesh (80%), Nagaland (75%), Bihar (62%) and Goa (28%) as depicted in figure 3.

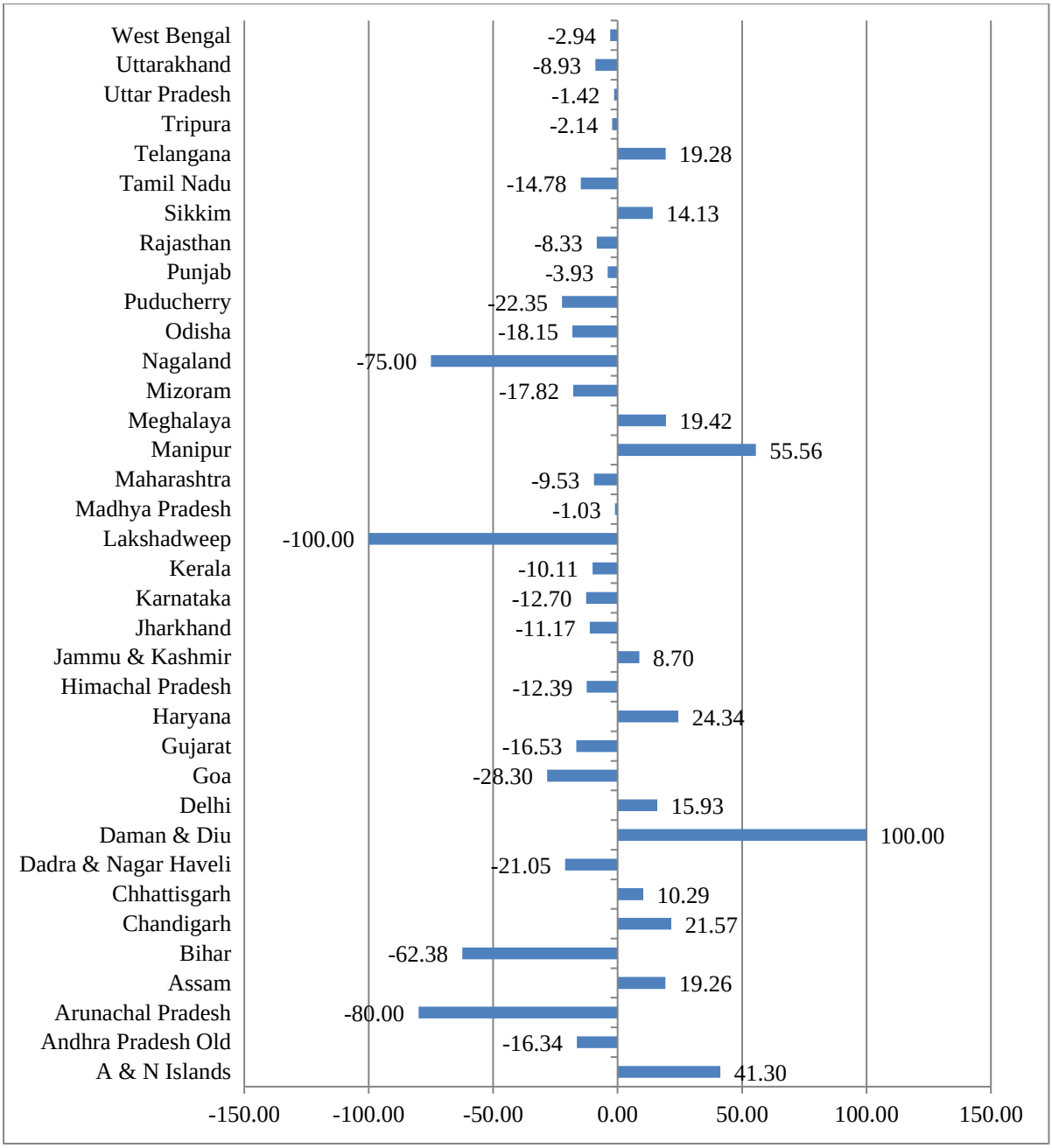

Figure 3: Suicide deaths growth rate, 2019 over 2018, HMIS

2.3. Suicide deaths in India: Trends and patterns in top 5 states

The HMIS data gives the number of suicides of 6 years and above age group of all health facilities in India. The few Indian states have the burden of more than 60 % of suicides in the country and those falling under top 5 states from 2010 to 2019 is depicted in Table 1 & Table 2. In 2010, Andhra Pradesh has the highest number of suicide reported which is 22 % followed by Karnataka (21%), West Bengal (11%), Tamil Nadu (10%), and Madhya Pradesh (8%) respectively. In this year the combined percentage share of suicides of top five states was 71 %.

Table 1: Top five Indian states with respect to suicide (HMIS – 2010 to 2014)

S. No.	States	2010	2011	2012	2013	2014
1	Andhra Pradesh	1st (21.9%)	2nd (14.1%)	2nd (16.4%)	2nd (18.1%)	4th (9.1%)
2	Karnataka	2nd (21.3%)	1st (27.1%)	1st (24.9%)	1st (20.9%)	1st (16.8%)
3	West Bengal	3rd (10.6%)	4th (7.0%)	4th (7.2%)	4th (9.4%)	3rd (9.6%)
4	Tamil Nadu	4th (10.1%)	NA	NA	5th (7.5%)	5th (8.9%)
5	Madhya Pradesh	5th (7.5%)	NA	NA	NA	NA
6	Maharashtra	NA	3rd (11.8)	3rd (16.1%)	3rd (14.5%)	2nd (16.4%)
7	Chhattisgarh	NA	5th (6.2%)	5th (6.2%)	NA	NA
8	Telangana	NA	NA	NA	NA	NA
	Total	**71.5%**	**66.2%**	**70.8%**	**70.5%**	**60.8%**

Source – HMIS Data

Table 2: Top five Indian states with respect to suicide (HMIS - 2015 to 2019)

S. No.	States	2015	2016	2017	2018	2019
1	Andhra Pradesh	4th (9.3%)	5th (8.7%)	5th (8.2%)	NA	NA
2	Karnataka	1st (15.7%)	2nd (14.0%)	4th (11.6%)	3rd (11.9%)	3rd (11.1%)
3	West Bengal	5th (8.5%)	NA	3rd (11.6%)	4th (10.6%)	4th (10.9%)
4	Tamil Nadu	3rd (13.4%)	1st (17.2%)	1st (16.2%)	1st (14.9%)	1st (13.5%)
5	Madhya Pradesh	NA	NA	NA	5th (8.4%)	5th (8.8%)
6	Maharashtra	2nd (15.6%)	3rd (13.9%)	2nd (13.1%)	2nd (12.7%)	2nd (12.2%)
7	Chhattisgarh	NA	NA	NA	NA	NA
8	Telangana	NA	4th (9.1%)	NA	NA	NA
	Total	**62.6%**	**62.9%**	**60.7%**	**58.6%**	**56.6%**

Source – HMIS data

In 2011, Karnataka had the highest number of suicides recorded from the past year (27%) followed by Andhra Pradesh (14%), Maharashtra (12%), West Bengal (7%), and Chhattisgarh (6%). The combined percentage share of suicides was 66 % in 2011. In 2012, Karnataka again had recorded the

highest number of suicides (25%), followed by Andhra Pradesh (16%), Maharashtra (16%), West Bengal (7%), and Chhattisgarh (6%) respectively.

The total percentage share of suicides for all the top 5 states were 71 % in 2012. In 2013, Karnataka again recorded highest number of suicides (21%) followed by Andhra Pradesh (18%), Maharashtra (14%), West Bengal (9%), and Tamil Nadu (8%) respectively. Total percentage share of suicides of all top five states in 2013 was 70 %. In 2014, Karnataka has the highest number of suicides (17%) followed by Maharashtra (16%), West Bengal (10%), Andhra Pradesh (9%), and Tamil Nadu (9%) respectively. The total percentage share of suicide in 2014 for above mentioned states was 61%. In 2015, Karnataka had the highest number of suicides (16%) followed by Maharashtra (16%), Tamil Nadu (13%), Andhra Pradesh (9%), and West Bengal (9%) respectively. Total percentage share of suicides in 2015 of all five states mentioned above was 63 %. In 2016, Tamil Nadu had the highest number of suicides recorded (17%) followed by Karnataka (14%), Maharashtra (14%), Telangana (9%), and Andhra Pradesh (9%) respectively. Total percentage share of suicides of top five states in 2016 was 63 %. In 2017, Tamil Nadu had the highest number of suicides recorded (16%) followed by Maharashtra (13%), West Bengal (12%), Karnataka (12%), and Andhra Pradesh (8%) respectively. The total share of the suicides of all the top five states in 2017 was 61 %. In 2018, Tamil Nadu had the highest suicide recorded. (15%) followed by Maharashtra (13%), Karnataka (12%), West Bengal (11%), and Madhya Pradesh (8%) respectively. The percentage share of the suicide of all the above-mentioned states in 2018 was 59 %. In 2019, Tamil Nadu again had the highest suicide recorded i.e. (14%) followed by Maharashtra (12%), Karnataka (12%), West Bengal (11%), and Madhya Pradesh (9%) respectively. The total percentage share of suicides of all five states mentioned above in 2019 was 57 %. The total number of suicides in all the Indian states over last one decade is presented in Table 3.

Table 3: Number of Suicides (6 years and above) in Indian states from 2010-2019 (HMIS)

States	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
All India	17678	26679	27476	30956	37248	45565	48701	46691	50894	47882
A & N Islands	66	45	47	35	52	50	63	49	46	65
Andhra Pradesh Old	3883	3769	4509	5610	3406	4238	4228	3820	3403	2847
Arunachal Pradesh	5	6	3	5	4	3	8	17	20	4
Assam	270	482	716	773	737	658	671	893	971	1158
Bihar	32	66	155	68	1	32	75	91	101	38
Chandigarh	85	86	122	27	10	79	83	71	51	62

Chhattisgarh	110	1641	1697	2105	2168	2483	2267	2086	2119	2337
Dadra & Nagar Haveli	38	60	64	73	80	88	86	62	76	60
Daman & Diu	7	21	17	18	9	11	10	12	8	16
Delhi	77	76	83	55	60	96	72	139	703	815
Goa			36	3	40	42	43	53	53	38
Gujarat	1119	1275	1043	1190	1465	1563	1448	2319	2347	1959
Haryana	159	249	224	430	816	1310	1581	388	1405	1747
Himachal Pradesh	338	387	375	371	384	401	381	305	444	389
Jammu & Kashmir	23	14	23	20	12	35	27	26	23	25
Jharkhand	82	98	199	184	180	163	137	175	206	183
Karnataka	3762	7217	6836	6496	6251	7165	6822	5419	6088	5315
Kerala	129	91	72	37	116	53	25	606	465	418
Lakshadweep	**0**	0	1	1	0	5	0	0	3	0
Madhya Pradesh	1332	1925	1369	1175	2110	2434	2458	4050	4267	4223
Maharashtra	149	3153	4418	4473	6116	7123	6811	6119	6452	5837
Manipur	37	34	40	35	55	57	52	47	45	70
Meghalaya	62	75	110	152	142	178	168	163	139	166
Mizoram	88	53	84	85	116	120	108	85	101	83
Nagaland	0	0	1	4	2	4	1	1	4	1
Odisha	530	883	543	635	846	1463	1521	1349	1339	1096
Puducherry	86	94	71	120	48	37	56	115	85	66
Punjab	229	485	478	315	696	840	966	1433	1451	1394
Rajasthan	1015	1251	639	963	1061	1014	1056	602	2040	1870
Sikkim	22	7	20	21	37	67	49	107	92	105
Tamil Nadu	1792	1049	1273	2334	3299	6124	8357	7563	7598	6475
Telangana	NA	NA	NA	NA	3007	3119	4414	2171	2375	2833

Tripura	75	88	115	82	116	184	180	224	234	229
Uttar Pradesh	149	81	72	97	119	344	338	549	565	557
Uttarakhand	58	39	31	67	130	90	118	145	168	153
West Bengal	1869	1879	1990	2897	3557	3892	4021	5437	5407	5248

Source – HMIS data

2.4. Suicide rates in India in the purview of NCRB data

NCRB is an Indian government agency responsible for collecting and analyzing the crime data as defined by Indian penal code and local laws. In this study, the suicide data is taken from the year 2010 to 2017. As per the NCRB data the suicide rate in the last few years shows a decreasing trend. In 2010 the total suicides were 134599 in India but in 2017 it decreased to 129887 which was a decrease of 3.5 % in the total number of suicide.

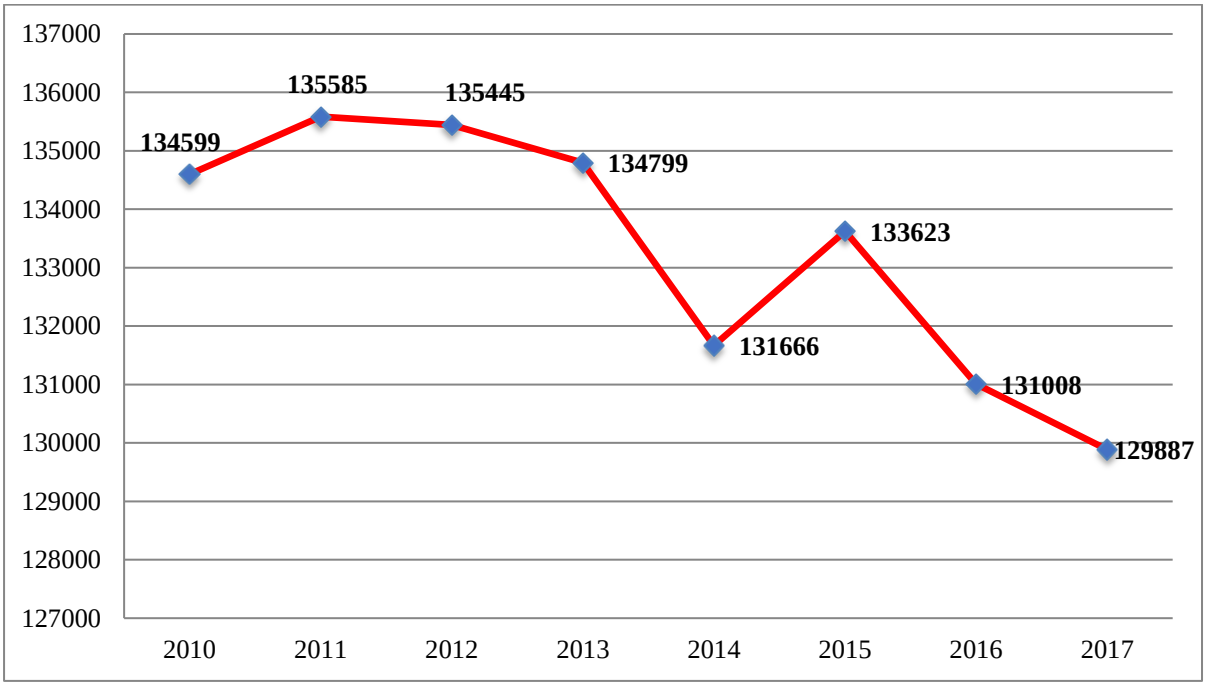

Figure 4: Number of suicides in India 2010-2017 (NCRB)

2.5. Suicide Annual Growth Rate

The states and UTs which have reported significant increase in suicides in 2011 over 2010 were Nagaland (from 12 in 2010 to 33 in 2011 i.e. an increase of 175%) followed by Chandigarh (48%), Meghalaya (42%), Uttar Pradesh (34%), Odisha (23%), Uttarakhand (13%), Haryana (12%), Delhi (11%) and Jammu & Kashmir (11%) as compared to the national average of 1%. Highest number of decrease was reported in Bihar (from 1,226 in 2010 to 795 in 2011) (Figure 5).

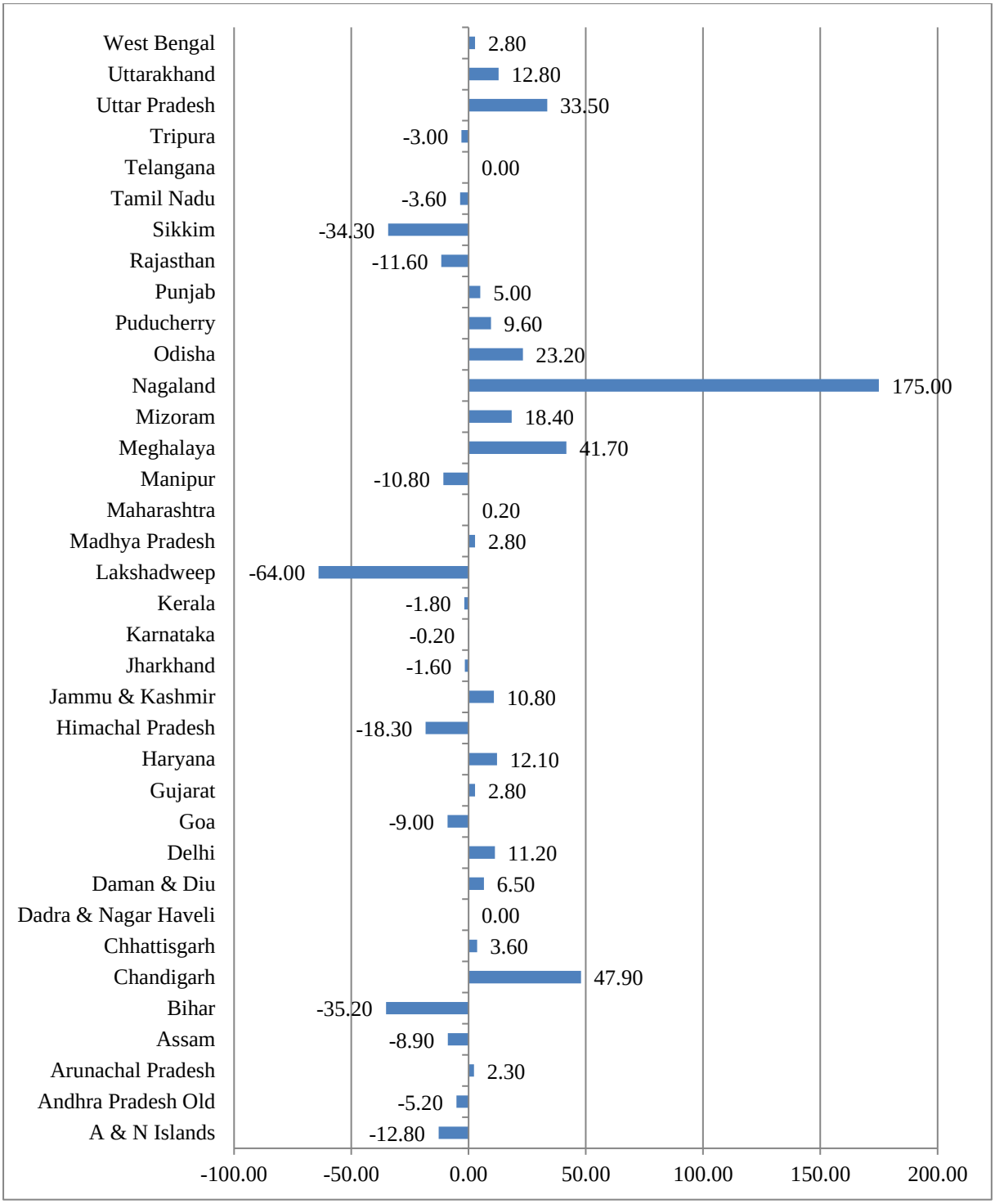

Figure 5: Suicide deaths growth rate, 2011 over 2010, NCRB

Similarly, in 2017 over 2016, the highest suicidal growth rates came from Uttarakhand (118%), Dadra & Nagar Haveli (36%), Uttar Pradesh (26%), Manipur (16%) and Odisha (15%) while highest percentage decrease was reported in Lakshadweep (100%), Mizoram (27%), Arunachal Pradesh (26%), Assam (25%) and Sikkim (19%) as depicted in figure 6.

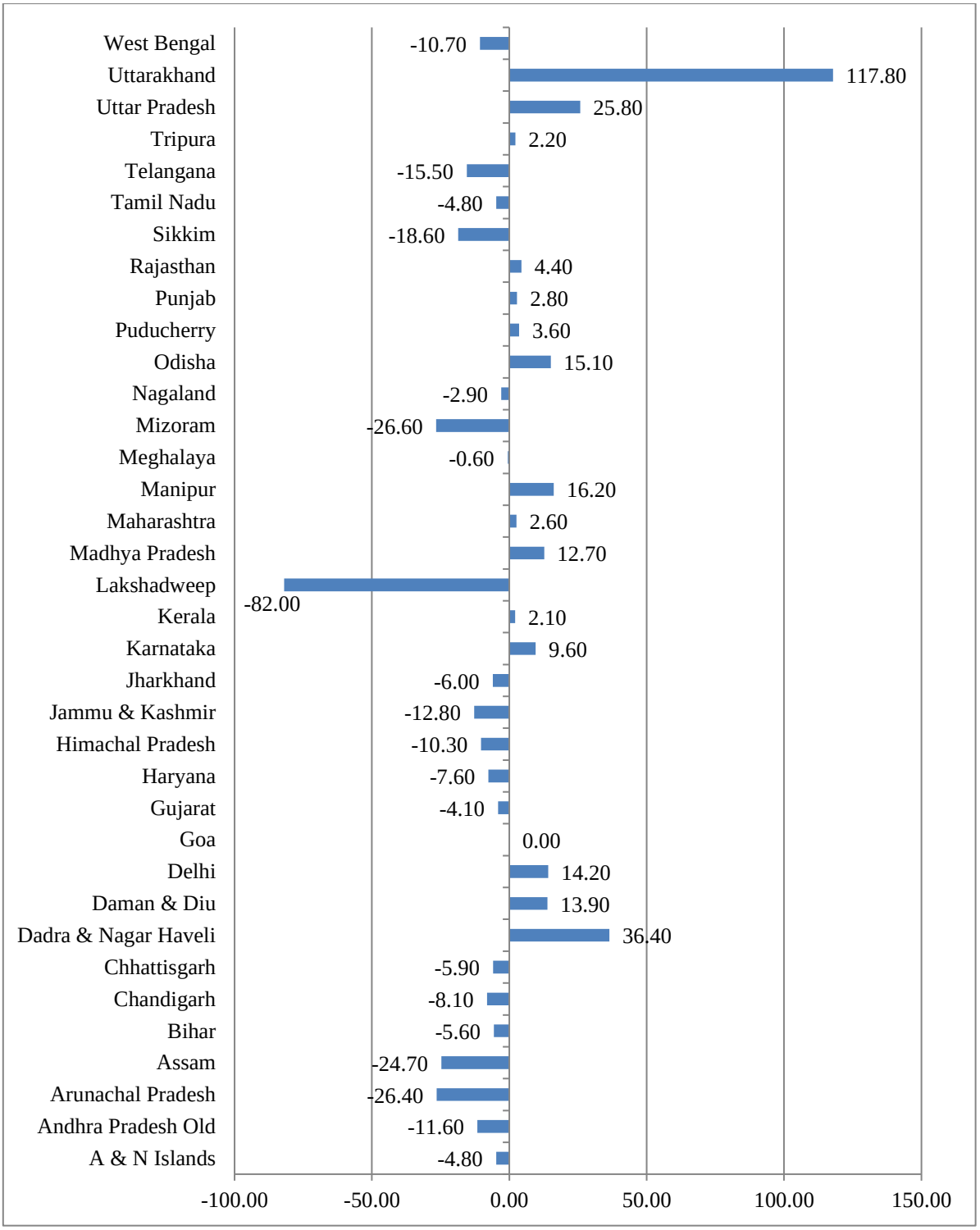

Figure 6: Suicide deaths growth rate, 2017 over 2016, NCRB

2.6. Suicide deaths in top 5 states of India

As per the NCRB report, few states contribute to more than 50 % of the total suicides in India. In 2010, Tamil Nadu recorded the highest number of suicide (12%) followed by West Bengal (12%), Andhra Pradesh (12%), Maharashtra (12%), and Karnataka (9%) respectively. The total percentage

share of suicides of above-mentioned states in 2017 was 57 %. In 2011, West Bengal recorded highest number of suicides (12%) followed by Maharashtra (12%), Tamil Nadu (12%), Andhra Pradesh (11%), and Karnataka (9%) respectively. The total percentage share of suicide of above-mentioned states in 2011 was 56 %.

Table 4: Top five Indian states with respect to suicide (NCRB – 2010 to 2013)

S. No.	States	2010	2011	2012	2013
1	Andhra Pradesh	3rd (11.8%)	4th (11.1%)	4th (10%)	3rd (10.8%)
2	Karnataka	5th (9.4%)	5th (9.3%)	5th (9.4%)	5th (8.4)
3	West Bengal	2nd (11.9%)	1st (12.2%)	3rd (11%)	4th (9.7%)
4	Tamil Nadu	1st (12.30%)	3rd (11.8%)	1st (12.5%)	2nd (12.3%)
5	Madhya Pradesh	NA	NA	NA	NA
6	Maharashtra	4th (11.8%)	2nd (11.8%)	2nd (11.9%)	1st (12.3%)
7	Telangana	NA	NA	NA	NA
	Total %	**57.2**	**56.2**	**55.3**	**53.5**

Source – NCRB Data, NA – Not Applicable

Table 5: Top five Indian states with respect to suicide (NCRB – 2014 to 2017)

S.No.	States	2014	2015	2016	2017
1	Andhra Pradesh	NA	NA	NA	NA
2	Karnataka	4th (8.3%)	4th (8.1%)	4th (8.2%)	5th (9%)
3	West Bengal	3rd (10.9%)	3rd (10.9%)	3rd (10.3%)	3rd (9.2%)
4	Tamil Nadu	2nd (12.2%)	2nd (11.8%)	2nd (11.6%)	2nd (11.1%)
5	Madhya Pradesh	NA	5th (7.3%)	5th (8%)	4th (9.1%)
6	Maharashtra	1st (12.4%)	1st (12.7%)	1st (13.1%)	1st (13.6%)
7	Telangana	5th (7.3%)	NA	NA	NA
	Total %	**51.1**	**51.2**	**51.2**	**52**

Source – NCRB Data, NA – Not Applicable

In 2012, Tamil Nadu again recorded the highest number of suicides (13%) followed by Maharashtra (12%), West Bengal (11%), Andhra Pradesh (10%), and Karnataka (9%) respectively. The total percentage share of suicide of the above-mentioned states in 2012 was 55 %. In 2013, Maharashtra recorded the highest number of suicides (12%) followed by Tamil Nadu (12%), Andhra Pradesh (11%), West Bengal (10%), and Karnataka (8%) respectively. The total percentage share of suicides of above-mentioned states was 54 %. In 2014, Maharashtra recorded the highest number of suicides (12%) followed by Tamil Nadu (12%), West Bengal (11%), Karnataka (8%), and Telangana (7%) respectively. The total percentage share of suicides of top 5 states mentioned above in 2014 was 51 %. In 2015, Maharashtra recorded the highest number of suicides (13%) followed by Tamil Nadu (12%), West Bengal (11%), Karnataka (8%), and Madhya Pradesh (7%) respectively. The total percentage share of the suicides of the above-mentioned states in 2015 was 51 %. In 2016, Maharashtra recorded the highest number of suicides (13%) followed by Tamil Nadu (12%), West Bengal (10%), Karnataka (8%), and Madhya Pradesh (8%) respectively. The total percentage share of suicides of the above-mentioned states in 2016 was 51 %. In 2017, again Maharashtra recorded the highest number of suicides (14%) followed by Tamil Nadu (11%), West Bengal (9%), Madhya Pradesh (9%), and Karnataka (9%) respectively. The total percentage share of suicides of all the above-mentioned states in 2017 was 52 %. All the top five Indian states with respect to highest number of suicides is shown in Table 4 and Table 5.

3. Comparative analysis of suicides from HMIS & NCRB

For this study, suicide data is taken from the two sources i.e. HMIS and NCRB. Both the sources are the government data. HMIS data is a collective data coming from all India's public health facilities however the NCRB is dedicated in collection and analyzing the crime data all over the India. Suicide data from the HMIS does not include all the aspect that NCRB includes in its data. NCRB suicide data covers various components like age, sex, reasons of suicide, mode of suicide etc. But HMIS gives only the suicide number who has minimum 6 years and above. HMIS has advantage that its data is monthly updated at the block level to national level but NCRB compiles the data annually and its repository have the latest data as on 2017 only.

3.1. Annual trend of suicides: HMIS vs NCRB

Figure 6 represents the suicide trends from HMIS and NCRB data. There is a difference between HMIS and NCRB data about suicides. HMIS suicide data shows the total number of suicides over the years increasing but in NCRB suicide data it is slightly decreasing. The reason behind the difference in figure 7 arises because of both the source belongs from the two different arens and their data is coming from multiple sources. HMIS suicide data has fixed sources which are coming from the public health facilities all over India.

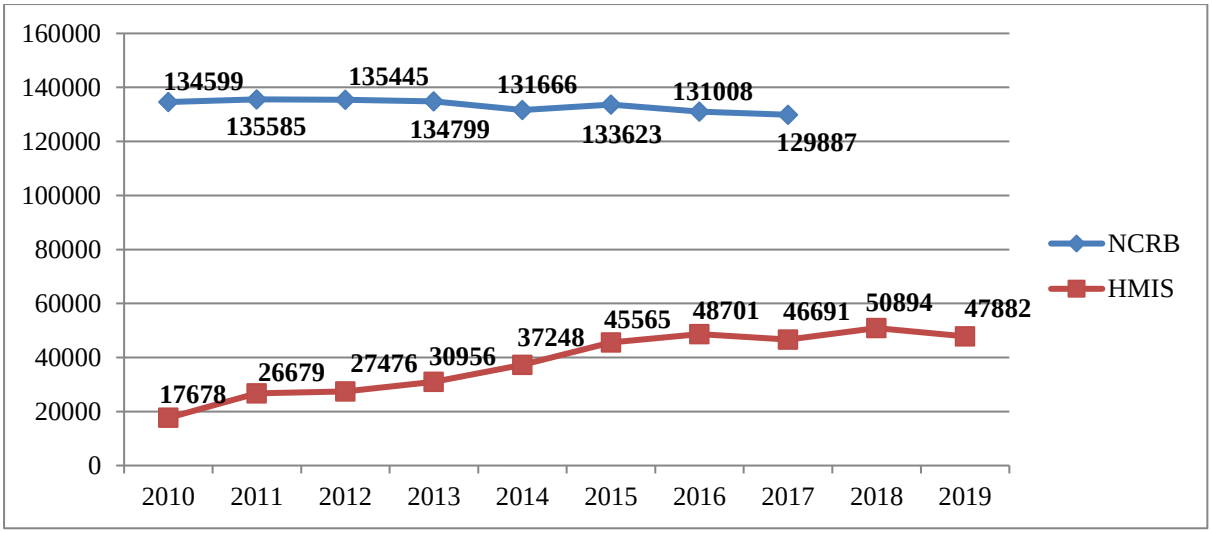

Figure 5: Suicides trends HMIS vs NCRB

3.2. Annual Suicide Growth Rate: HMIS vs NCRB

Annual suicide growth rate is calculated for HMIS (Figure 2 & 3) and NCRB (Figure 4 & 5) for their respective years. The growth rate of year 2011 over 2010 for HMIS shows the drastic increase of suicide rates in Maharashtra and Chhattisgarh which were 2016 and 1392 % respectively. But as per NCRB data, Nagaland and Chandigarh had the highest increase i.e. 175 % and 47 % respectively. The growth rate of year 2019 over 2018 of HMIS shows 100 % and 55.56 % increase in Daman & Diu and Manipur respectively. In this year HMIS suicide data shows more negative growth rates rather than positive growth rate which is a good sign. The growth rate of NCRB of 2017 over 2016 shows Uttarakhand has the highest increase 118 % followed by Dadar Nagar Haveli 36 %.

3.3. Top 5 states with respect to suicides: HMIS vs NCRB

Top five states according to HMIS and NCRB are same except Chhattisgarh which got the 5th place in year 2011 and 2012 continuously according to HMIS data. Rests of the other states show similar state level ranks position of both datasets comprising of Andhra Pradesh, Maharashtra, West Bengal, Karnataka, Tamil Nadu, Madhya Pradesh and Telangana. The rankings and the total percentage share of suicides for HMIS data is shown in Table 1 and Table 2. Similarly, the ranking and total percentage share of the states for NCRB suicide data is shown in Table 4 and Table 5.

3.4. Availability of Mental Health Workers to curb suicides in India

For implementation of any national program in India or anywhere in the world require infrastructure and Human Resources. For the NMHP the number of Psychologist & Psychiatrist plays a major role for assessing and treatment of different type of mental problem. The total number of Psychiatrists sanctioned and posted in different states of India was taken from the HMIS portal depicted in table 6.

The number of sanctioned post of Psychiatrists in top five states of India that have the highest number of reported suicides Maharashtra (63), Andhra Pradesh (62), Tamil Nadu (43), West Bengal (41), and Karnataka (30) respectively. The number of posted Psychiatrist are 53 in Maharashtra, 51 in Andhra Pradesh, 35 in Tamil Nadu, 50 in West Bengal and 38 in Karnataka respectively. West Bengal and Karnataka have more number of Psychiatrists with respect to total number of sanctioned posts as per the 2019-20 HMIS data.

Table 6: Status of sanctioned and posted Psychiatrists in India (HMIS - 2019 to 2020)

S. No.	States	In position	Sanctioned	S. No.	States	In position	Sanctioned
1	Andhra Pradesh	51	62	**19**	Nagaland	NA	NA
2	Arunachal Pradesh	4	0	**20**	Odisha	1	1
3	Assam	25	0	**21**	Punjab	14	5
4	Bihar	2	2	**22**	Rajasthan	8	12
5	Chhattisgarh	1	3	**23**	Sikkim	NA	NA
6	Goa	0	0	**24**	Tamil Nadu	35	43
7	Gujarat	41	46	**25**	Telangana	12	14
8	Haryana	2	3	**26**	Tripura	6	0
9	Himachal Pradesh	8	14	**27**	Uttar Pradesh	11	18
10	Jammu & Kashmir	0	0	**28**	Uttarakhand	0	1
11	Jharkhand	NA	NA	**29**	West Bengal	50	41
12	Karnataka	38	30	**30**	A & N Islands	1	1
13	Kerala	18	20	**31**	Chandigarh	22	24
14	Madhya Pradesh	10	12	**32**	Dadra & Nagar Haveli	1	1
15	Maharashtra	53	63	**33**	Daman & Diu	NA	NA
16	Manipur	NA	NA	**34**	Delhi	35	74

17	Meghalaya	1	0	**35**	Lakshadweep	NA	NA
18	Mizoram	0	0	**36**	Puducherry	1	1

Source – HMIS, NA- Data not Available

3.5. Budget Approved under NMHP for top 5 suicide prevalent states

Every year, each district's budget needs under various national programmes are requested through Programme Implementation Program (PIP) and the budget approval report by the government is send in the form of Record of Proceedings (ROP). Table 7 depicts the budget approved under NMHP for top 5 Indian states for year 2015 to 2019. In this budget summary, the amount is added for some major heads of NMHP, which are community based service delivery, community interventions, infrastructure, procurement, referral transport, training, IEC/BCC, and Public Private Partnership (PPP). It is not compulsory that the demanded and approved budget would constitute all the major heads because these budgets are demanded according to their requirements. In the last five years from 2015 to 2019, West Bengal got the highest budget allocated (7.6 Cr) under NMHP followed by Karnataka (4.5Cr.), Maharashtra (2.2 Cr.), Tamil Nadu (2.0 Cr.), and Andhra Pradesh (2.0 Cr.)

Table 7: Year wise Approved Budget for NMHP (2015-2019)

Year	Karnataka	Maharashtra	Tamil Nadu	Andhra Pradesh	West Bengal
2019-20	96100000	81140000	30650000	61100000	10926000
2018-19	96100000	0	39700000	45500000	26400000
2017-18	67598000	22570000	19763000	NA	NA
2016-17	159811000	18792000	66959000	93775000	36257000
2015-16	31720000	97600000	43480000	0	2400000
Total	**451329000**	**220102000**	**200552000**	**200375000**	**75983000**

Source – ROP, NA – Data not available

3.6. An overview of reasons and age compositions for suicides in India

There could be a various reason of suicide and most of the data did not classify the exact reasons for the suicides. According to NCRB these kinds of data consist more than 20 % of suicides where reasons of suicides are not known. As per the latest NCRB suicide data (2017) 'Family Problems' and 'Illness' were the major causes of suicides which accounted for 30% and 19% of total suicides respectively during 2017. 'Marriage Related Issues' (6%), 'Drug Abuse/Addiction' (5%) 'Bankruptcy or Indebtedness' (4%), 'Love Affairs' (4%), 'Failure in Examination' (2%), 'Unemployment' (2%),

‘Professional/Career Problem’ (1%) and ‘Property Dispute’ (1%) were other causes of suicides. The reason of suicides in the age group up to 18 years was highest (2251) due to family problem followed by failure in exam (1489), causes not known (1431), other causes (1384), illness which includes AIDS, Cancer, Paralysis, Insanity/Mental Illness and other prolonged illness (1113) and love affairs (1006) respectively. In 18 to 30 years age group, the highest number of suicides recorded was due to family problems (13973) followed by illness (5802), cause not known (5497), other causes (5469), and love affairs (3317) respectively.

Table 8: Age-specific Reasons for suicides in India (2017)

Reasons	up to 18 years	18 - 30 years	30 - 45 years	45 - 49 years	60 & above
Bankruptcy or Sudden change in Economic Status	35	900	2184	1618	414
Suspected/Illicit Relation	25	315	324	107	3
Cancellation/Non-Settlement of Marriage	77	1134	651	105	5
Not having Children(Barrenness/Impotency)	9	143	150	51	10
Illness	1113	5802	6895	5977	4201
(i) AIDS/STD	2	43	91	67	14
(ii) Cancer	11	144	362	381	277
(iii) Paralysis	11	104	277	388	248
(iv) Insanity/Mental Illness	599	3012	3128	2241	1266
(v) Other Prolonged Illness	490	2499	3037	2900	2396
Death of Dear Person	67	307	324	228	148
Dowry Dispute	28	1501	525	51	1
Divorce	4	184	204	85	1
Drug Abuse/Addiction	50	1726	2693	1759	477

Failure in Examination	1489	984	62	4	1
Fall in Social Reputation	25	190	216	137	39
Family Problems	2251	13973	13096	7241	2477
Ideological Causes/Hero Worshipping	13	36	21	19	5
Illegitimate Pregnancy	5	20	17	0	0
Love Affairs	1006	3317	692	83	2
Physical Abuse (Rape, Incest etc.)	33	46	24	1	3
Poverty	35	339	461	262	101
Professional/Career Problem	74	636	629	434	102
Property Dispute	48	296	459	392	106
Unemployment	37	1152	842	307	66
Causes Not known	1431	5497	4550	2197	804
Other Causes	1384	5469	4955	2894	1149
Extra Marital Affairs	17	458	445	82	15
Marriage related Others issues	60	792	522	231	18

Source – NCRB Data

In 30 to 45 years age groups the highest number of suicides recorded was due to family problems (13096), followed by illness (6895), other causes (4955), cause not known (4550), drug abuse/addiction (2693), and bankruptcy or sudden change in economic status (2184). In 45 to 49 years age group, the highest number of suicides were reported due to family problems (7241), followed by illness (5977), other causes (2894), cause not known (2197), drug abuse/addiction (1759), and bankruptcy or sudden change in economic status (1618) respectively. In 60 years and above age groups the highest number of suicides recorded was due to illness (4201), followed by family problem (2477), other causes (1149), causes not known (804), and drug abuse/addiction (477) respectively. Figure 8 depicts the major reasons of suicides in India which consists of 80 % of suicides reasons in 2018.

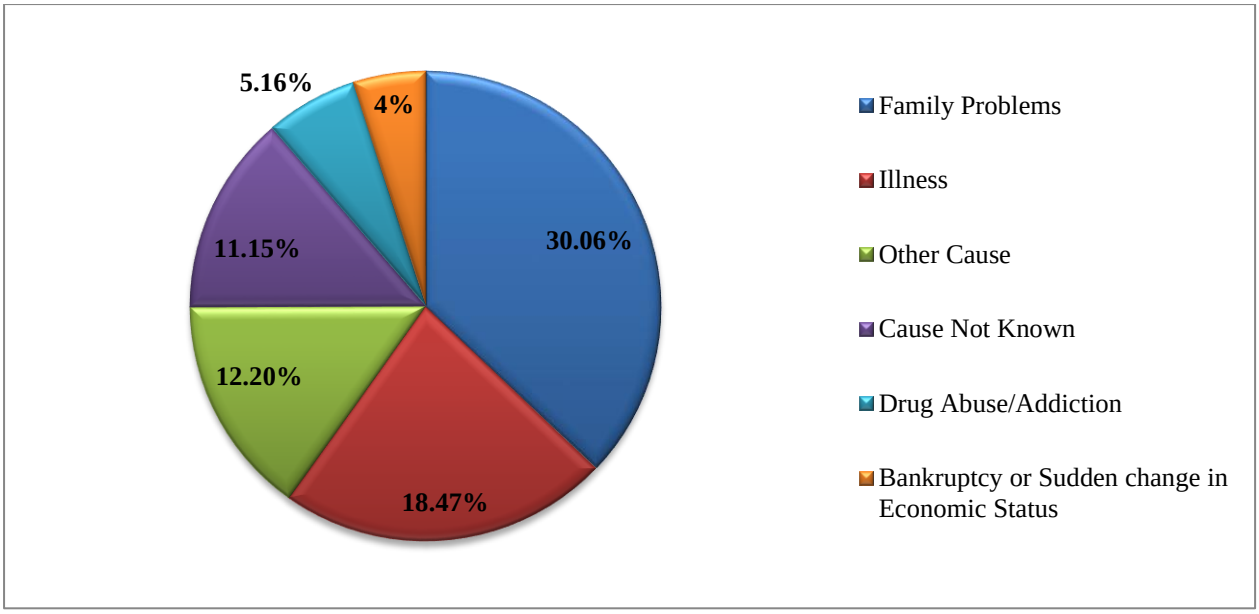

Figure 6: Major reasons of suicides in India (NCRB 2017)

3.7. Distribution of Suicidal Cases as per Sex and Age Composition

The visual summary which shows in figure 9 (a – h) depicts the distribution of suicidal cases and age wise composition over the time period 2010-2017. It is clearly visible that from 2010 onwards there has been slight increase in suicidal cases as per sex and age wise composition. From 2010, the maximum numbers of suicidal cases were observed in the age group of 15-29 years with a total of 47625 or approximately 35 % with 26387 males and 21238 females' suicides. It is followed by 44846 suicidal cases in the age group of 30-44 years wherein 30444 males and 14402 females' suicide reported in the same year. The major cause behind the suicide in these two particular age groups were illness, family problems, causes which were unknown, and other causes which were reported for their suicides. On the contrary, the least suicidal cases were seen in less than 14 years of age contributing to only 2 % of suicide rates but the maximum suicides were done by males than females respectively. However, there has been a decline in the suicide statistics from 2010 to 2012 with 134599 in the former year and 120488 in the latter year. There has been a decline in 11 % in these two years with similar age composition of suicides of males and females. Maximum number of suicides reported to be around 41793 in 15-29 years of age and least suicides have been reported (2314 suicides) in the age group upto 14 years. The major reason behind such suicides are family problems, followed by illness specifically insanity and prolonged illness among both males and females in India. From 2014 onwards, another fact became observable with respect to transgenders who too were suiciding with 13 number of suicides being done in the age group of 18-30 though main cause behind the suicide is still unknown. In the year 2015, the number of males have been more than females on an aggregate level with 91528 males, 42088 females, and 7 transgenders who committed suicides respectively.

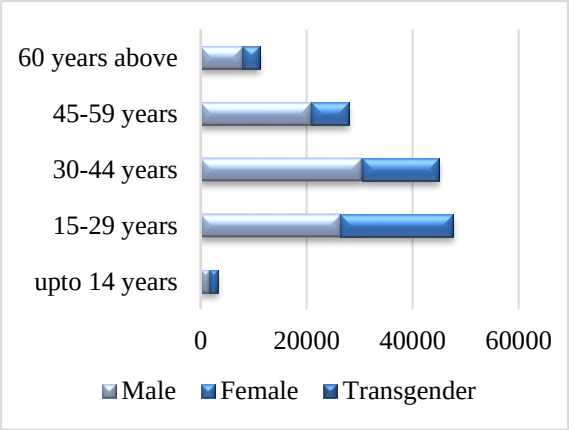

Figure 7(a): Age & Gender (NCRB – 2010)

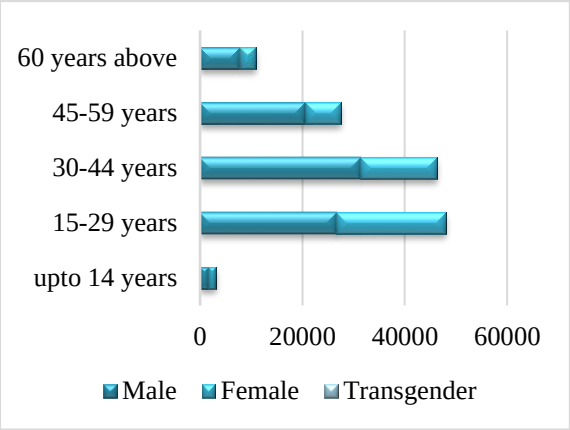

Figure 9(b): Age & Gender (NCRB – 2011)

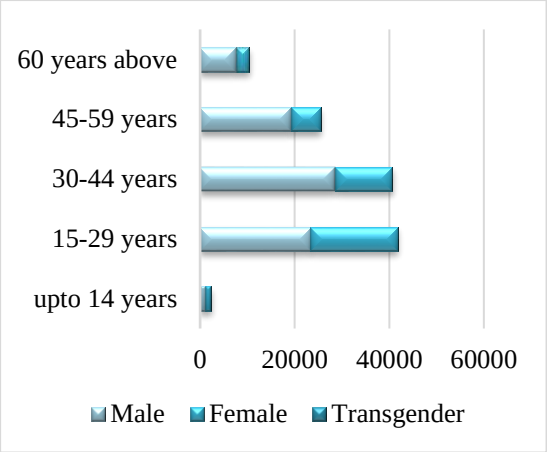

Figure 9(c): Age & Gender (NCRB – 2012)

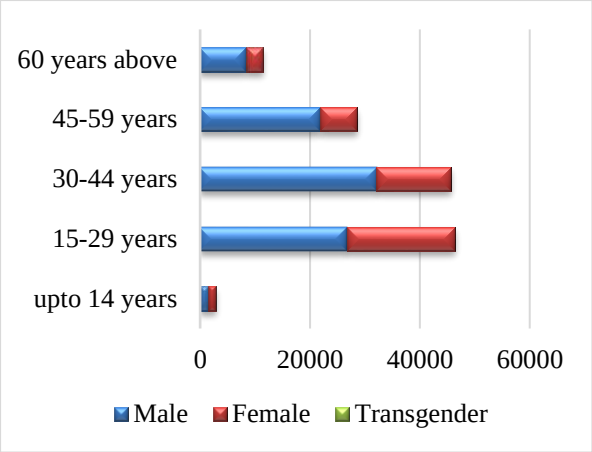

Figure 9(d): Age & Gender (NCRB – 2013)

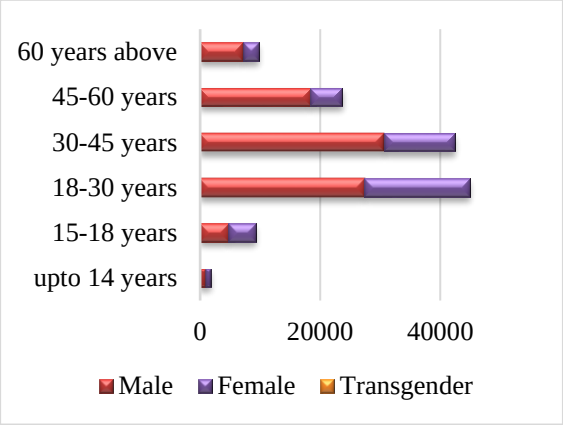

Figure 9(e): Age & Gender (NCRB – 2014)

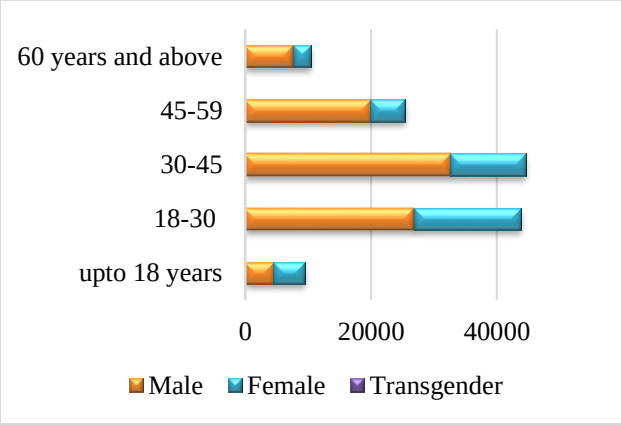

Figure 9(f): Age & Gender (NCRB – 2015)

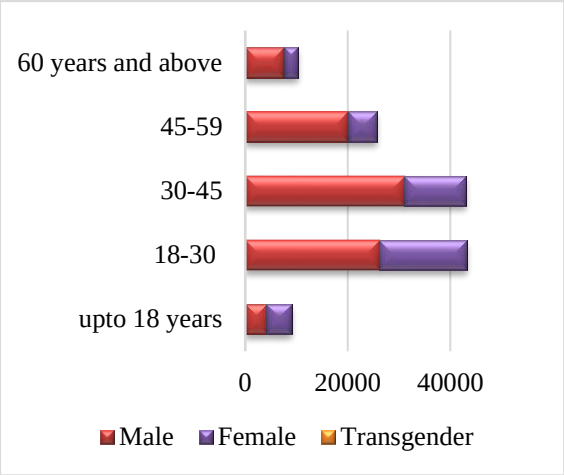

Figure 9(g): Age & Gender (NCRB – 2016)

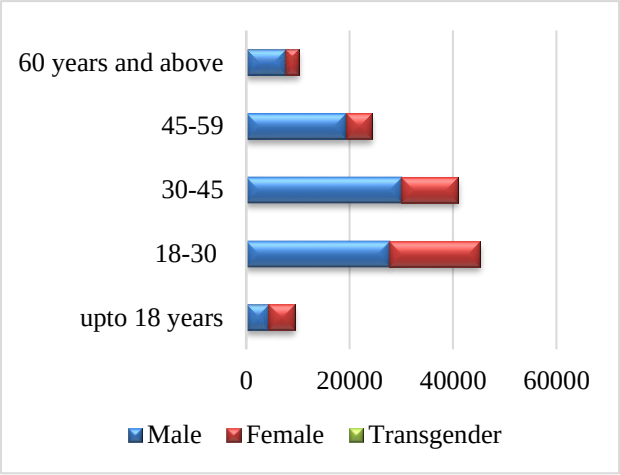

Figure 9(h): Age & Gender (NCRB – 2017)

4. Implication for NMHP with respect to suicides in India

Suicidal deaths are a serious global health burden and its prevention requires an integrated, all-inclusive, multi-sectoral and holistic approach. Community and institutional measures to cater to the probable risk factors leading to suicide deaths must be strengthened. National Health Mission is the flagship programme of MoHFW aiming to provide "Health for all". As the meaning to health per se broadens, suitable interventions/guidelines will have to be adapted to the changing dynamics. The recent surge in mental health awareness must be tapped upon by states and key stakeholders all over and spreading awareness must be continued until the stigma around treatment seeking for mental health disorders is de-stigmatised. It is essential, given the recent, epidemiological shift, that the states strengthen the integration of mental health intervention and the "conventional" health service basket.

The country like India where major part of the country still thinks mental problem is not a health issue or a serious issue and because of this suicide become the major cause of deaths reported due to mental health problem in our country. More than one lakh people died due to suicide in India and this is only because of the less awareness and stigma related to mental health problem. To reduce the burden of mental health the Government of India introduced NMHP in 1982 and later DMHP in 1996 at district level. It could be further improved if this program will be introduced at community level properly. But the major barrier to this programme has been the reluctance and stigma related to mental health problem and treatment seeking thereafter. It has been noted and addressed by many district health officials, that under the NMHP special camps were organised but nobody comes for the screening/check-up. Lots of funds allocated in the name of NMHP were unutilized and carry forwarded to next year. As per the HMIS data the numbers of suicides are increasing in the last ten years. It can only be reduced by spreading awareness from city to village level and allocation and training of dedicated staff for counselling of mental health problem. Under NMHP a number of

dedicated hospitals have been upgraded. Sum of five lakhs were given to each NGOs under NMHP for IEC material, organising camps etc. Training of ANMs and ASHAs related to mental health problems could be good initiatives to reach the micro level. Women still can have counselling from ANM and ASHAs at village level but men are reluctant towards this problem and the major number of suicides are committed by males as compared to females (see figure 9 to 16). Adolescents can be screened at the Adolescent Friendly Health Clinics. Very limited data is available with respect to mental health and suicides in India. A couple of resources like HMIS and NCRB record the suicide data in India. A dedicated portal could be a way forward of monitoring and follow-ups for the patients like MCTS portal for maternal health. Self-help groups and 24 hrs suicide help line number for all over India can reduce the burden of suicide.

4.1. Good practices/ Innovation

Mental health is now part of comprehensive primary healthcare. It is also anticipated that the community would begin to use the Health and Wellness Centre (HWC) as the first point of contact with the Primary Health Centre (PHC) serving as the first referral point. In this context, innovations like mobile tele-mental healthcare service, mobile phone for mental health risk identification, referral, follow-up as well as for data management at HWC is feasible. PHCs can be equipped with essential psychotropic drugs and teleconferencing or video conferencing for specialist consultation by psychologists and psychiatrists on a fortnightly or monthly basis from District or State level healthcare facility. Using tele-mental health technology (NIMHANS ECHO model), district level general health, and mental health personnel can be trained periodically. The District Hospitals and Hospitals for Mental Health can be strengthened by collaborative care model (community-based care and facility-based care services) under district mental health program with a tele-psychiatric mobile van, adequate financial resource, and the training. These hospitals can be linked with civil service organizations (CSO) offering mental health care services at PHC and HWC. Such programming, complimented with focused information, education, and communication (IEC) activities to eliminate mental health stigma/discrimination and improve access to mental health services can aid the effective implementation of the DMHP.

5. Conclusion:

Suicide is a global burden with respect to the deaths around the globe and to overcome this problem various mental health programs are running all over the world. Globally around 8 lakhs people died due to suicides (WHO, 2019) and out of it more than one lakh people died in India. If we compare the suicide data collected by NCRB then the percentage share of India to the total suicides in the world is around 14 %. In this study we attempt to see the change in the number of suicides in last 9 years (2010-2019). The data was collected from HMIS and NCRB to see the suicide trend of India in past few years and further annual suicide growth rates was calculated for both the data. On the basis of

total number of suicides, the top 5 states had been ranked according to years for both the data sources. Comparative analyses have been done between HMIS and NCRB suicide data. From HMIS the number of Psychiatrists sanctioned and in position in all states of India was shown for the year 2019-20. Thereafter the funds approved in PIP under NMHP for the top 5 states (HMIS) which was ranked on the basis of highest number of suicides recorded was also analysed. An overview of major reasons of suicides and reason by age group and sex is also portrayed.

The suicide trend comparison shows a contradicting result between HMIS and NCRB data. HMIS show increasing trend for last 9 years and NCRB has shown a declining trend. Similarly, for annual suicide growth rate there is no similarity between HMIS and NCRB data, which is due to the limitation of HMIS suicide data collection. HMIS collects data only from the public health facility level but NCRB has various resources for collection of data. The comparative analysis of the top 5 states with respect to highest number of suicides is almost same for both the data. The top 5 states mainly contributed from 6 states namely Andhra Pradesh, Maharashtra, Karnataka, West Bengal, Tamil Nadu, and Madhya Pradesh, which accounts for more than 60 % of total suicide deaths in India. The HR status with respect to Psychiatrist is competitively high in the above-mentioned states as compared to other Indian states. West Bengal and Karnataka have a greater number of psychiatrists in position to the sanctioned post. From the NCRB data, the major reasons of the suicides are family problems, illness, other causes, causes not known, drug abuse/ addiction and bankruptcy or sudden change in economic position which accounts for more than 80 % of suicides. As per the NCRB data, from 2010 to 2014 & 2016 to 2017 the maximum numbers of deaths were recorded in the age group of 15 to 19 years age and for 2015 maximum number of deaths was reported in the age group of 30 to 45 years.

Mental health as a part of comprehensive primary healthcare (CPHC), can be useful for NMHP/DMHP to reduce the burden of suicides in India. If the PHCs and Health and Wellness Centres (HWCs) can be equipped with essential psychotropic drugs and teleconferencing or video conferencing for specialist consultation by psychologists and psychiatrists on a fortnightly or monthly basis from District or State level healthcare facility it can go a long way. Use of collaborative care model for the secondary facilities and designated Mental Health facilities can be strengthened under district mental health program with a tele-psychiatric mobile van, adequate financial resource, and training. When these attributes are coupled with focused information, education, and communication (IEC) activities it can help to eliminate mental health stigma/discrimination and improve access to mental health services.

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